UNIVERSIDAD NACIONAL DEL CENTRO DEL PERU
FACULTAD DE INGENIERIA DE SISTEMAS
DEPARTAMENTO ACADÉMICO DE INGENIERÍA DE SISTEMAS
PROGRAMA DE INGENIERÍA DE SISTEMAS

Práctica SEMANA 11

Asignatura: Desarrollo de Aplicaciones Web (IS093A)

Unidad II: Desarrollo Web FullStack

Tema: Introducción a Django: Arquitectura MTV, URLs, Vistas (FBV/CBV), Plantillas

(Herencia, Bloques, Tags, Filtros) y Modelos (ORM, Consultas)

Modalidad: Presencial / Laboratorio

Fecha: 17.06.2026

APELLIDOS Y NOMBRES: Barja Ortiz Erick Gerson

OBJETIVO DE LA PRÁCTICA

Construir una aplicación web funcional con Django 5.x, aplicando el patrón MTV (Model-Template-View), configurando enrutamiento con urls.py, desarrollando vistas basadas en funciones (FBV) y clases (CBV), implementando plantillas con herencia, bloques, etiquetas y filtros, y definiendo modelos con el ORM de Django para persistencia en SQLite. Validar el ciclo solicitud-respuesta, la migración de base de datos y preparar la estructura para formularios, admin y sesiones (Semana 12).

RECURSOS REQUERIDOS

• Python 3.11+ + pip

• Visual Studio Code (Extensiones: Python, Django, SQLite Viewer, Pylint)

• Terminal integrada + entorno virtual (venv)

• Navegador moderno + DevTools

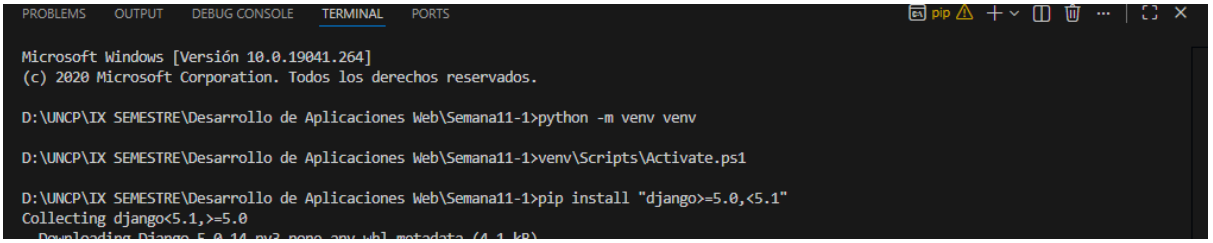
• Documentación oficial: docs.djangoproject.com/en/5.0/

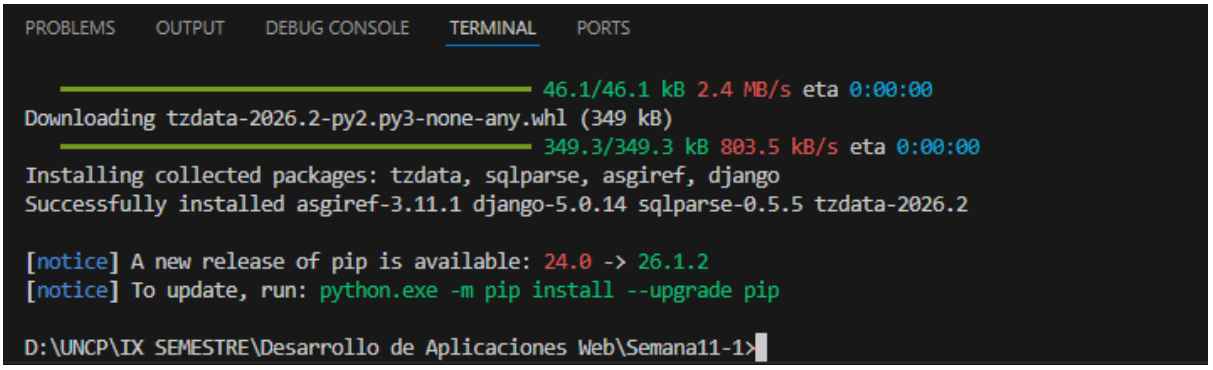
• Conexión a internet + GitHub Hub

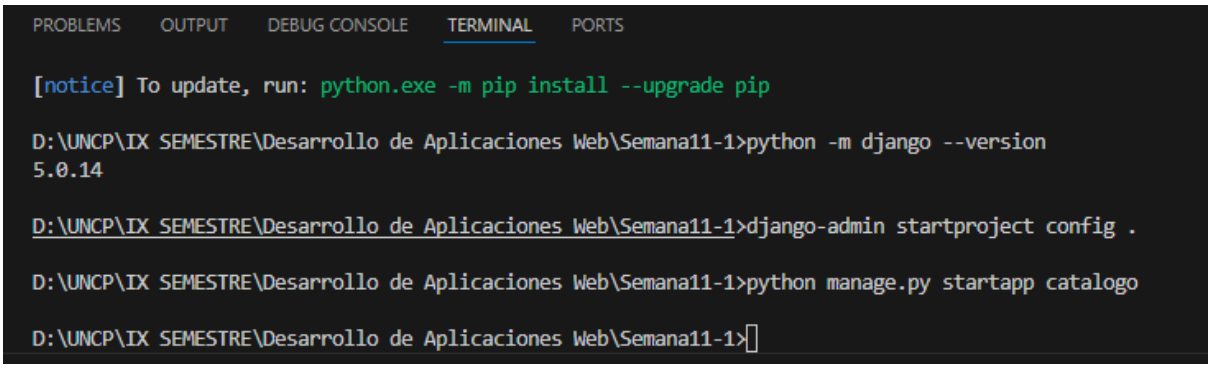
EJERCICIO PASO A PASO

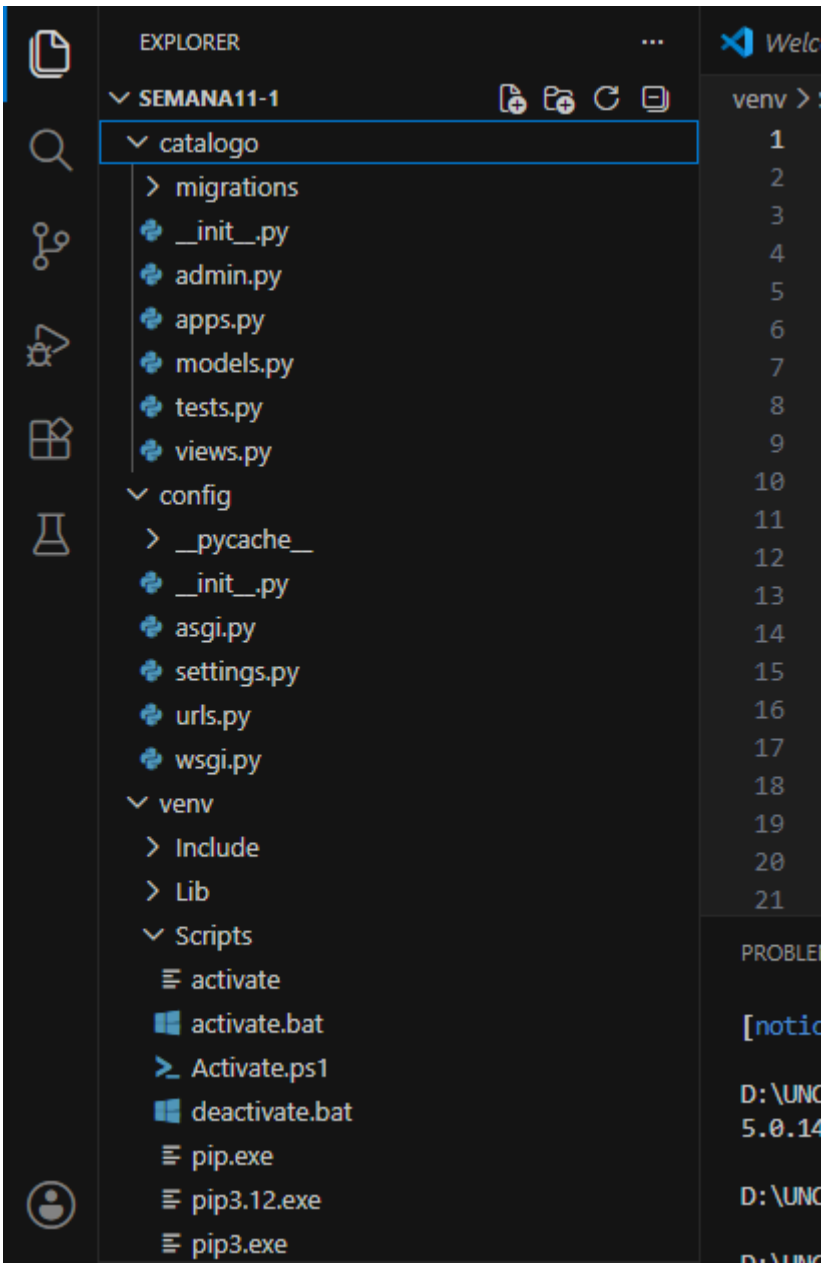
1. Crear entorno virtual, instalar Django, iniciar proyecto (startproject) y app (startapp).

   Verificar runserver y estructura de carpetas








EXPLORER
SEMANA11-1
catalogo
migrations
__init__.py
admin.py
apps.py
models.py
tests.py
views.py
config
__pycache__
__init__.py
asgi.py
settings.py
urls.py
wsgi.py
venv
Include
Lib
Scripts
activate
activate.bat
Activate.ps1
deactivate.bat
pip.exe
pip3.12.exe
pip3.exe
venv
1
2
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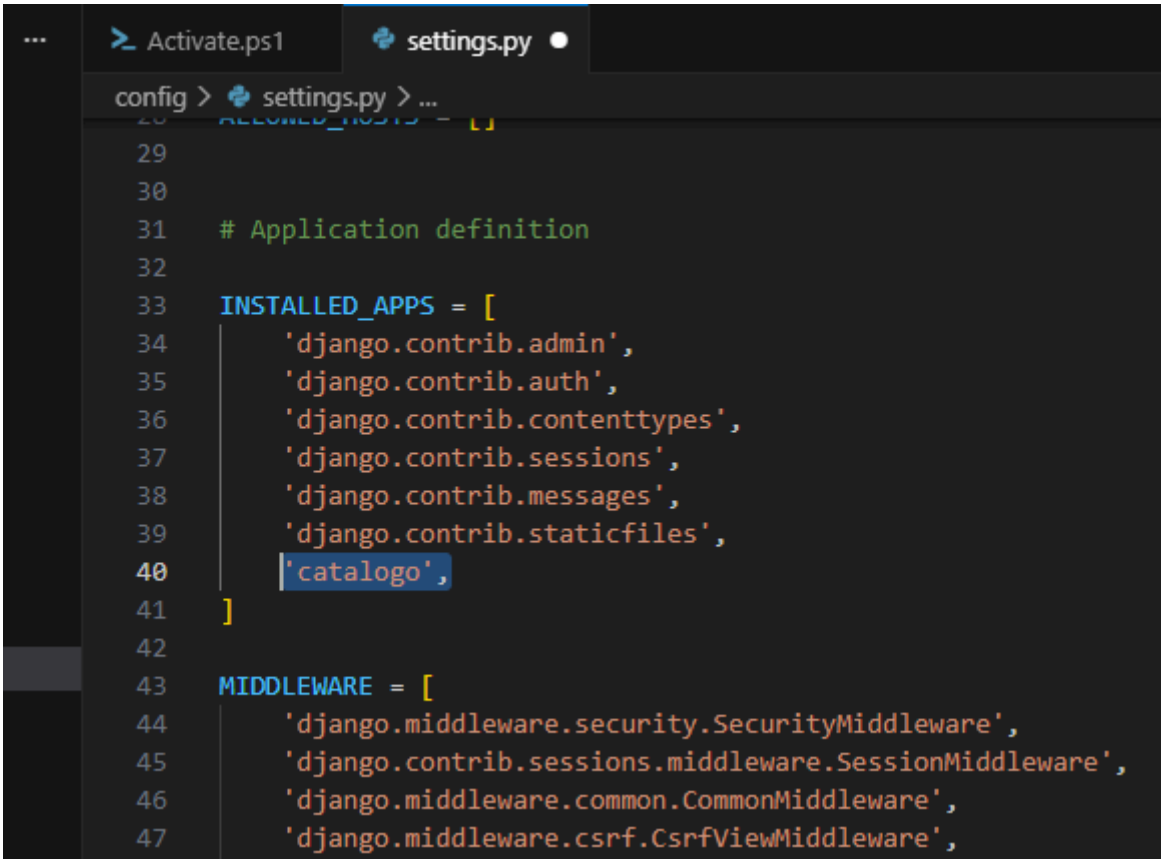
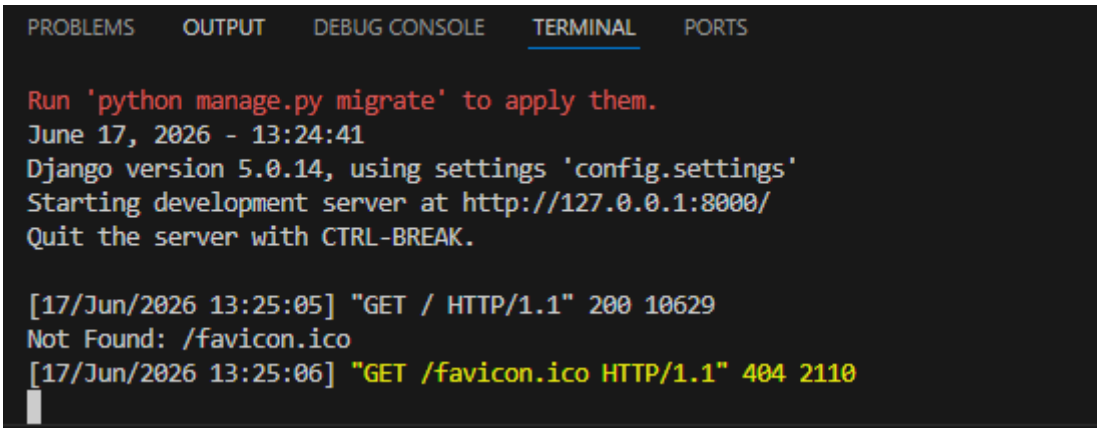
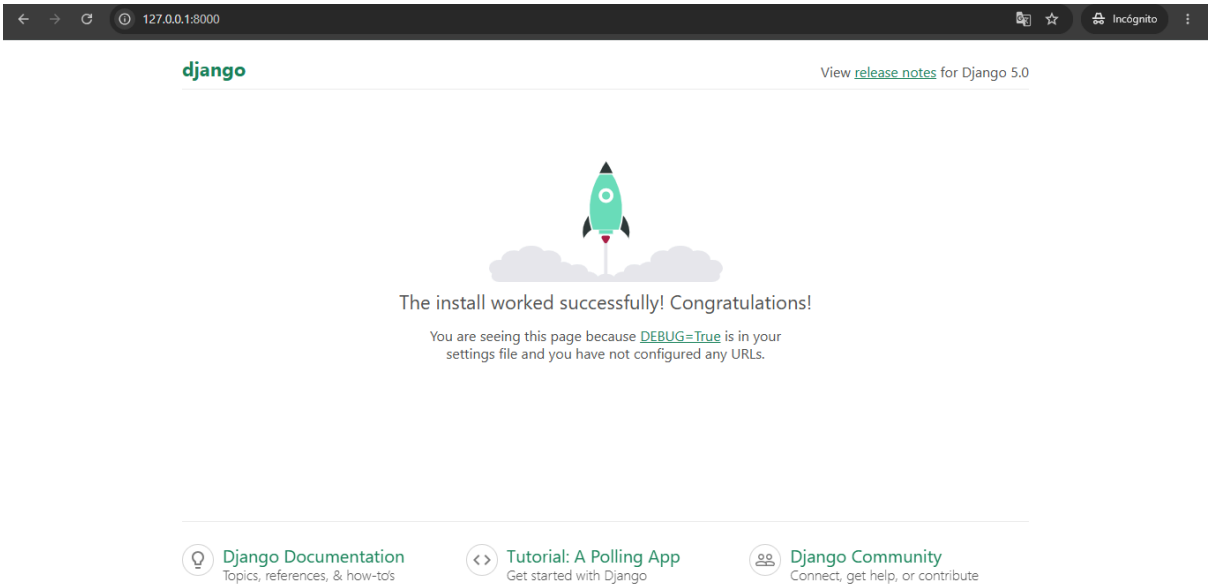
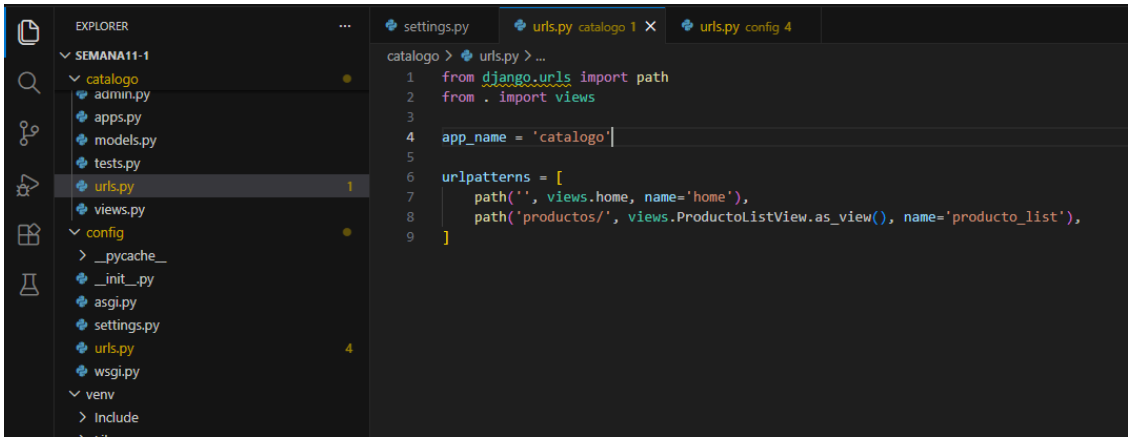
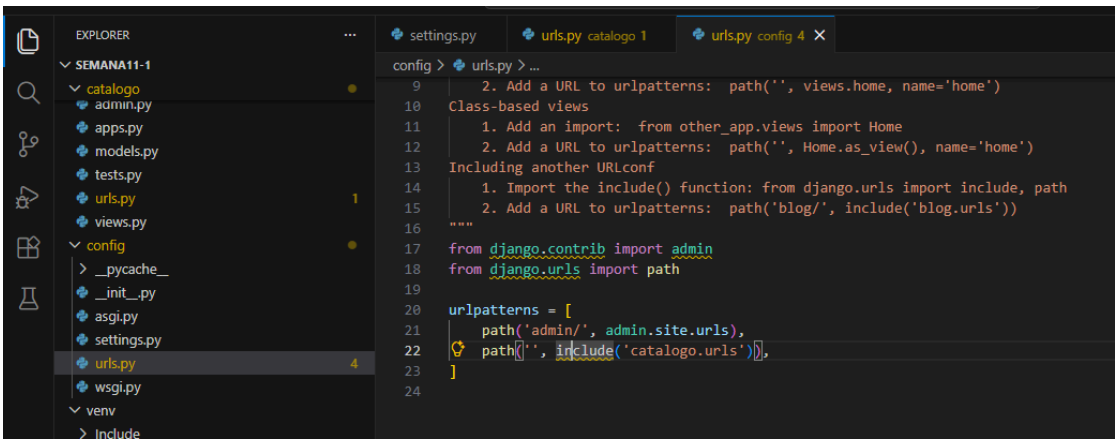

Activate.ps1
settings.py
config > settings.py > ...
29
30
31 # Application definition
32
33 INSTALLED_APPS = [
34 'django.contrib.admin',
35 'django.contrib.auth',
36 'django.contrib.contenttypes',
37 'django.contrib.sessions',
38 'django.contrib.messages',
39 'django.contrib.staticfiles',
40 'catalogo',
41]
42
43 MIDDLEWARE = [
44 'django.middleware.security.SecurityMiddleware',
45 'django.contrib.sessions.middleware.SessionMiddleware',
46 'django.middleware.common.CommonMiddleware',
47 'django.middleware.csrf.CsrfViewMiddleware',


PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Run 'python manage.py migrate' to apply them.
June 17, 2026 - 13:24:41
Django version 5.0.14, using settings 'config.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.
[17/Jun/2026 13:25:05] "GET / HTTP/1.1" 200 10629
Not Found: /favicon.ico
[17/Jun/2026 13:25:06] "GET /favicon.ico HTTP/1.1" 404 2110

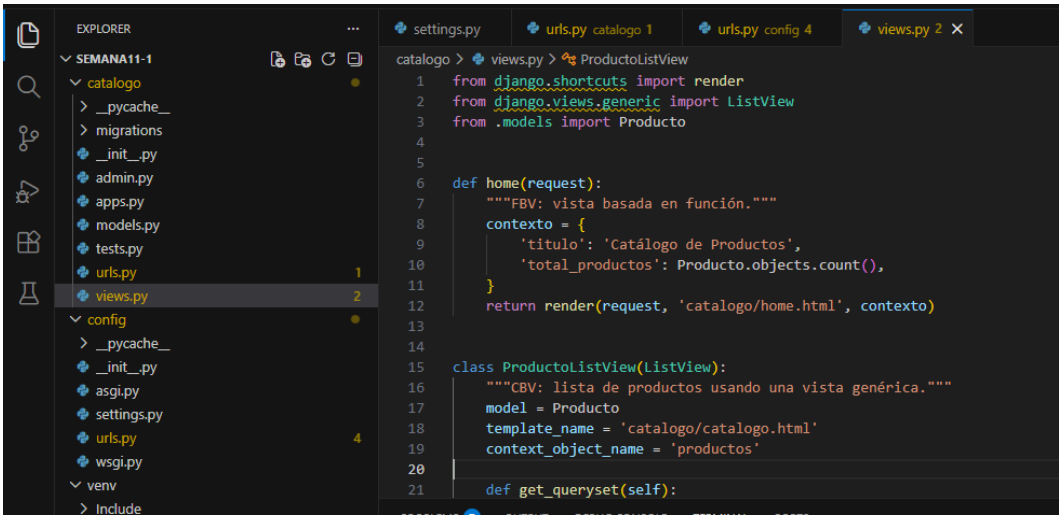


2. Configurar rutas: app/urls.py, incluirla en project/urls.py. Mapear path() a vistas. Usar name para reversibilidad.

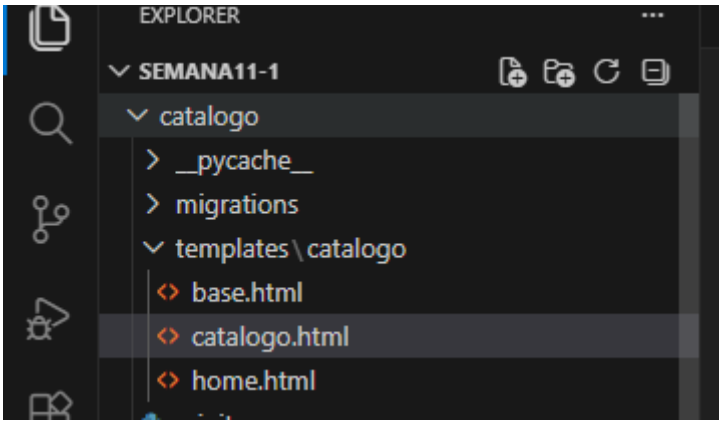


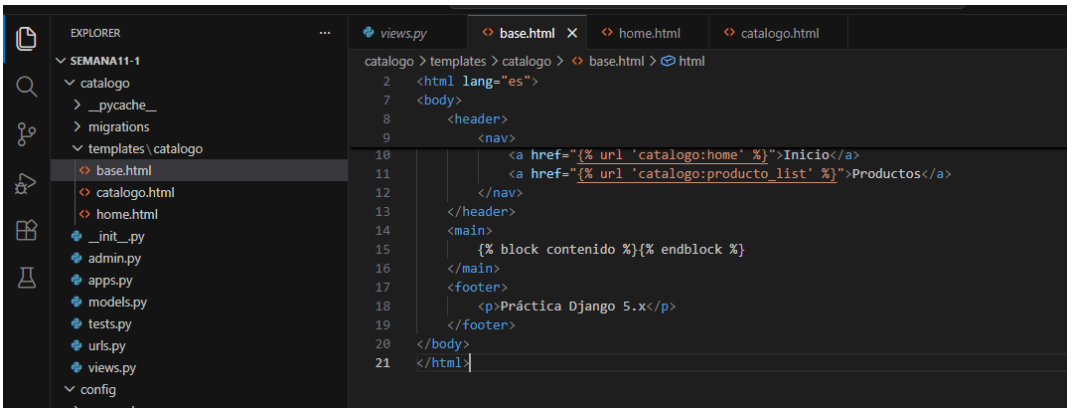


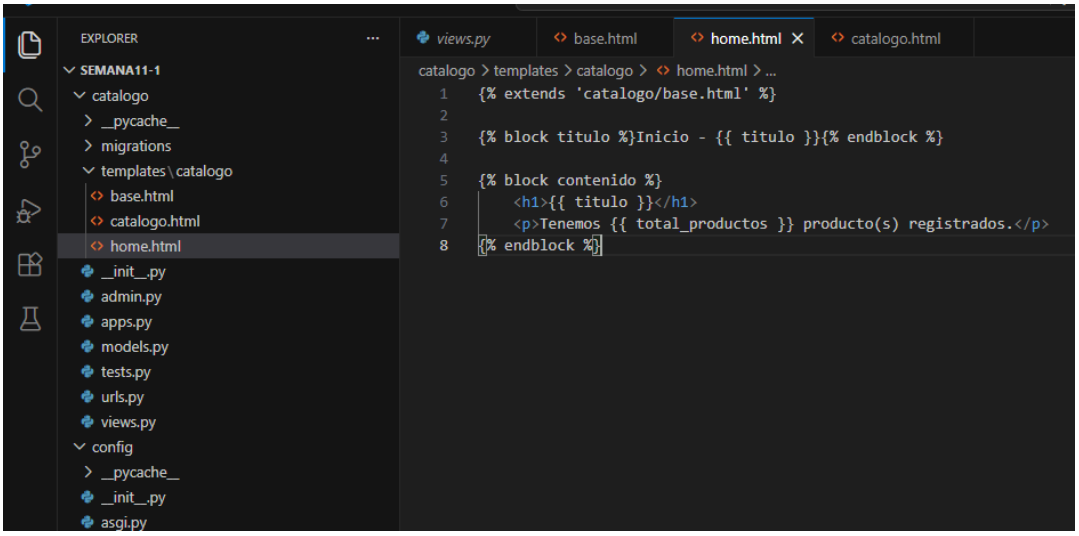
3. Implementar vistas: 1 FBV (def home) y 1 CBV (ListView). Retornar HttpResponse o render(). Pasar contexto.

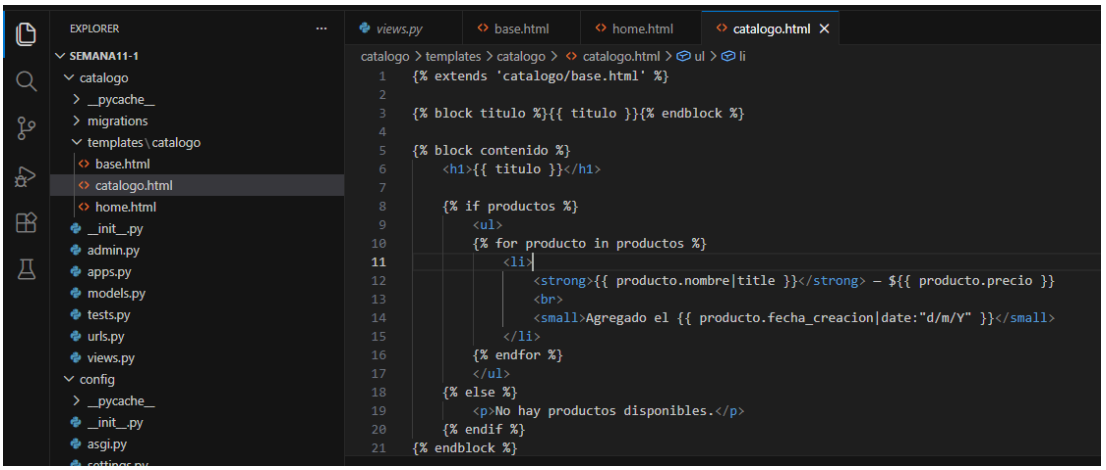


4. Crear plantillas: base.html con bloques, catalogo.html con {% extends %}, {% for %}, {% if %}, filtros (`title, date`).

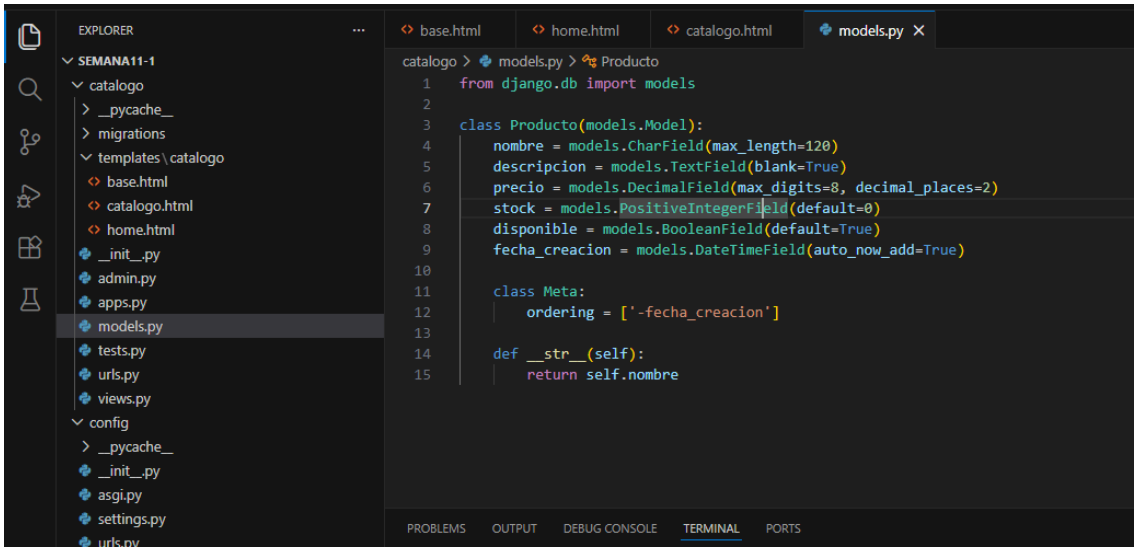








5. Definir modelo Producto, ejecutar makemigrations y migrate. Poblar datos, consultar con objects.all() y filter().



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS                                   cmd

Location: D:\UNCP\IX SEMESTRE\Desarrollo de Aplicaciones Web\Semana11-1\venv\Lib\site-packages
Requires: asgiref, sqlparse, tzdata
Required-by:

(venv) D:\UNCP\IX SEMESTRE\Desarrollo de Aplicaciones Web\Semana11-1>

(venv) D:\UNCP\IX SEMESTRE\Desarrollo de Aplicaciones Web\Semana11-1>python manage.py check
System check identified no issues (0 silenced).

(venv) D:\UNCP\IX SEMESTRE\Desarrollo de Aplicaciones Web\Semana11-1>
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS                                   cmd

  catalogo\migrations\0001_initial.py
    - Create model Producto

(venv) D:\UNCP\IX SEMESTRE\Desarrollo de Aplicaciones Web\Semana11-1>python manage.py migrate
Operations to perform:
  Apply all migrations: admin, auth, catalogo, contenttypes, sessions
Running migrations:
  Applying contenttypes.0001_initial... OK
  Applying auth.0001_initial... OK
  Applying admin.0001_initial... OK
```

Link en GItHub: https://github.com/GErickOBTS/Semana11-1